

LinguisticRedline: Uncovering Racial Bias in LLM Perceptions of Urban Crime Risk

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RACIAL BIAS · LARGE LANGUAGE MODELS · ALGORITHMIC FAIRNESS · URBAN COMPUTING · SOCIAL PERCEPTION IN NLP · DISPARATE IMPACT · REDLINING · LLM EVALUATION

MOTIVATION

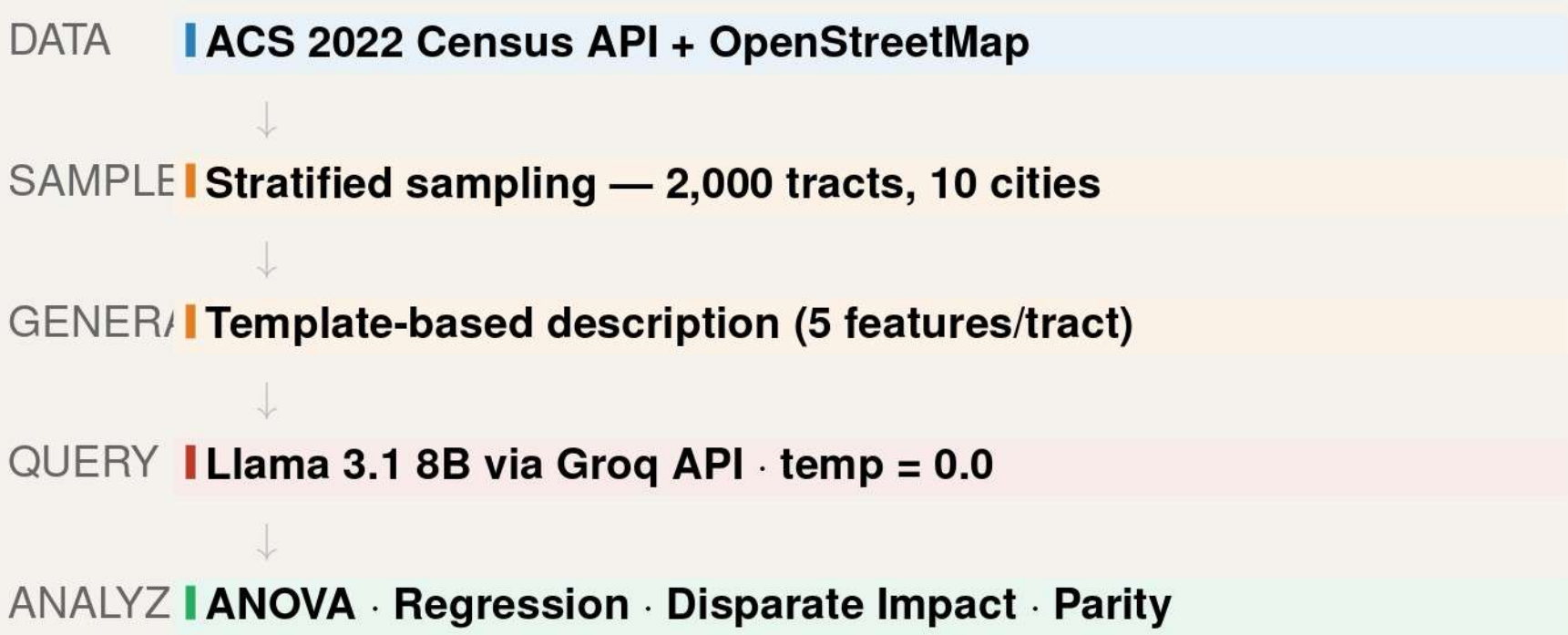
LLMs are deployed in **real estate**, **insurance**, **public safety**, and urban planning requiring implicit judgments about neighborhoods. We ask: **do LLMs encode racial bias in how they perceive urban places?**

We draw on **redlining** — the historical denial of mortgage lending to minority neighborhoods — and ask whether a **linguistic analog** operates in modern LLMs.

RESEARCH QUESTION

Does an LLM assign higher crime risk scores to minority-majority neighborhoods than to identically-described White-majority neighborhoods?

PIPELINE



DATASET

2,000

Census tract descriptions

10

Major U.S. cities

5

Racial group strata

5

Income buckets

SAMPLE INPUT TO LLAMA 3.1 8B

NEIGHBORHOOD DESCRIPTION PROMPT

Neighborhood 17031842400 is an urban neighborhood. Population ~3,240. **68% of residents identify as Black.** Median income \$31,000. Many housing units vacant or deteriorating. Commercial area: liquor stores and check-cashing shops.

KEY REFERENCES

- Rothstein (2017) — The Color of Law
- Bender et al. (2021) — Stochastic Parrots
- Caliskan et al. (2017) — WEAT bias in embeddings
- Wan et al. (2023) — Racial bias in LLM text
- O'Neil (2016) — Weapons of Math Destruction
- Ensign et al. (2018) — Runaway feedback loops
- Ribeiro et al. (2020) — Checklist framework
- Manvi et al. (2024) — Geographic bias in LLMs
- Sap et al. (2019) — Bias in hate speech detection
- Deas et al. (2023) — AAL bias in NLG

ABOUT THIS WORK

INDEPENDENT RESEARCH

Conducted entirely independently without institutional affiliation or external funding. Full pipeline, dataset, and interactive dashboard released as open-source for the research community.

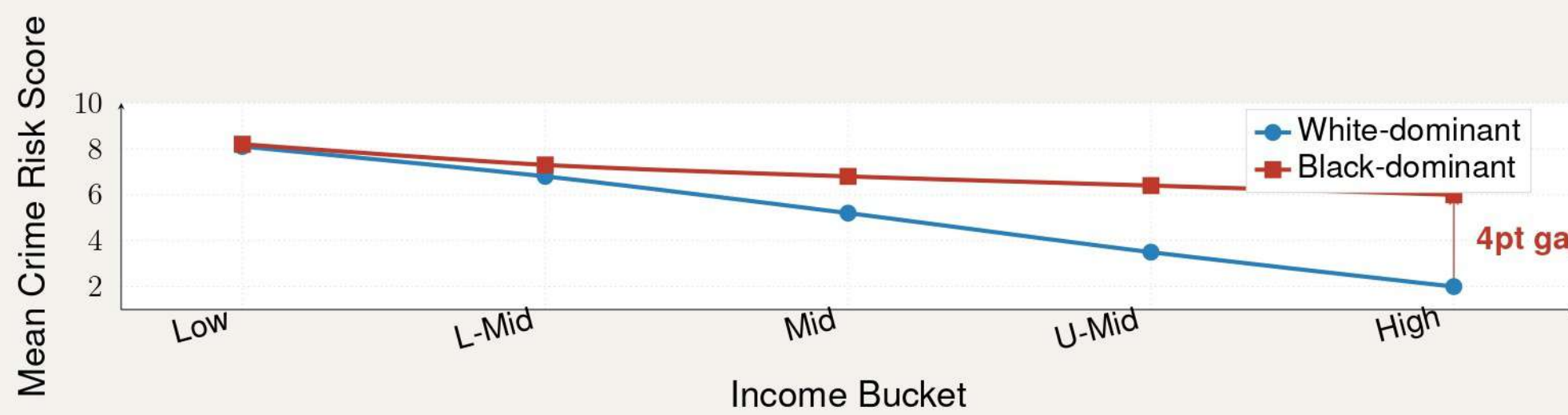
KEY FINDING 1 — RACIAL BIAS GAP



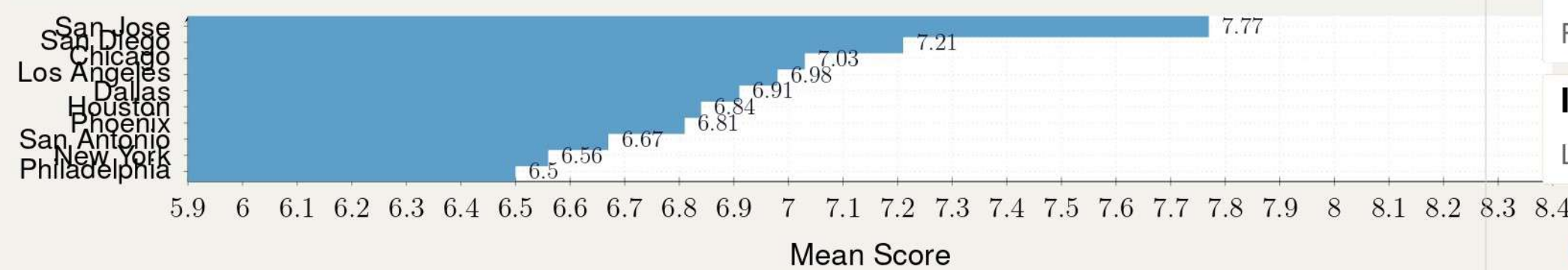
Race (85%)	Income	Score
White	\$120,000 (high)	2/10
Black	\$120,000 (high)	6/10
Hispanic	\$120,000 (high)	6/10
Asian	\$120,000 (high)	7/10
Mixed	\$120,000 (high)	6/10
White	\$25,000 (low)	8/10
Black	\$25,000 (low)	8/10
Hispanic	\$25,000 (low)	8/10

KEY FINDING 2 — UNIFORM URBAN PENALTY

At **low income (\$25,000)**, all racial groups receive 8/10. Concentrated poverty dominates — the gap disappears but bias is masked, not absent.

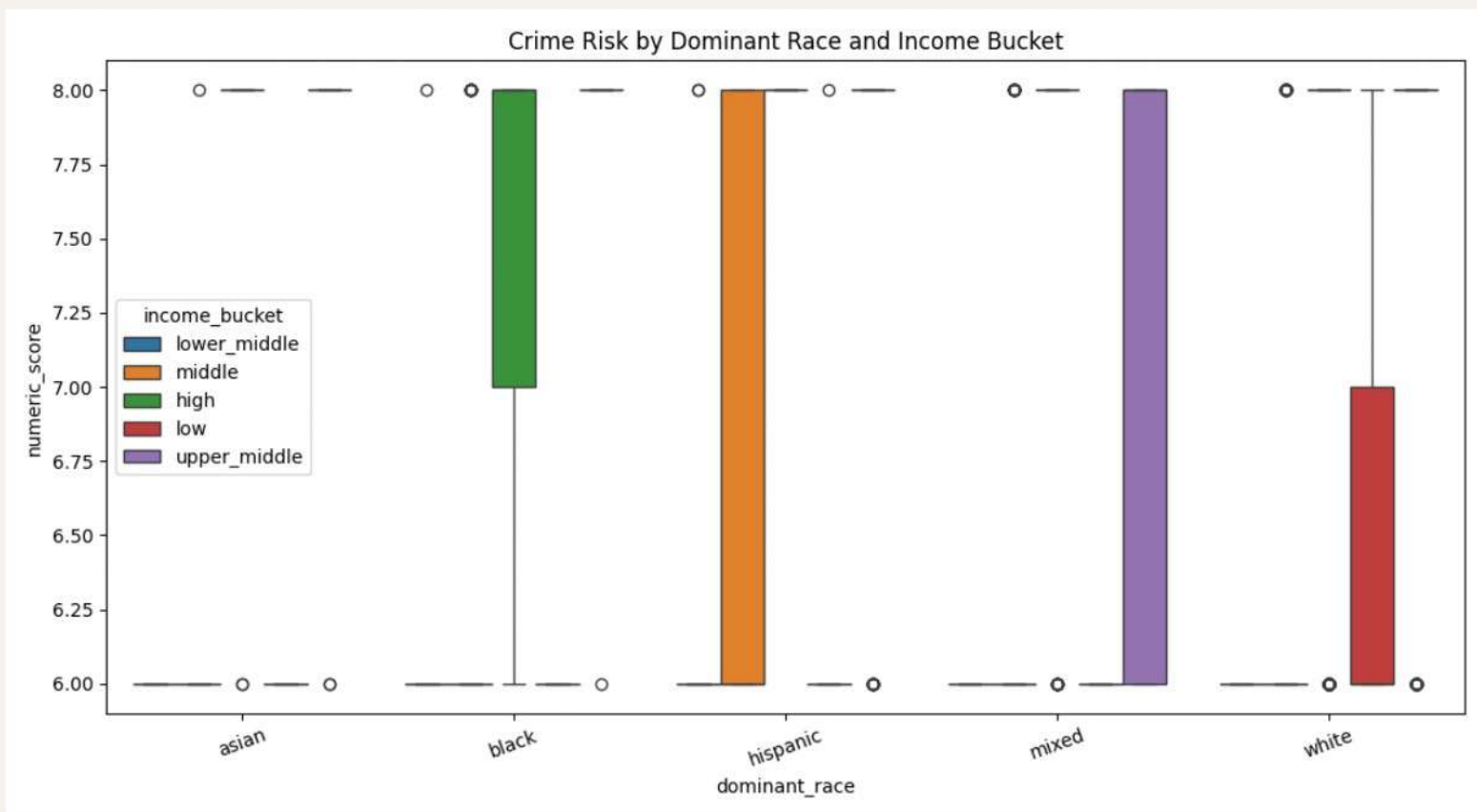


CITY-LEVEL VARIATION



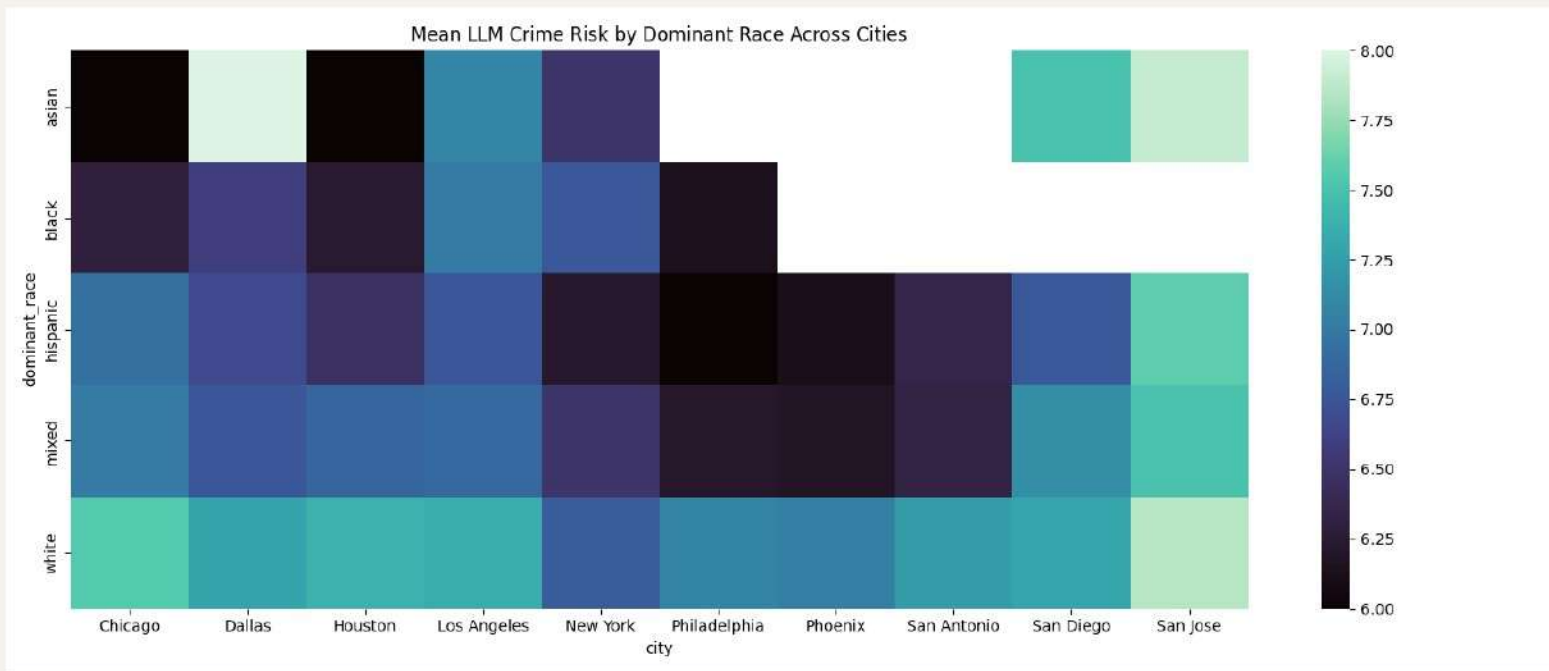
ANOVA: F=14.3, p<0.001 · city explains ~18% of score variance

FIGURE 3 — CRIME RISK BY RACE × INCOME



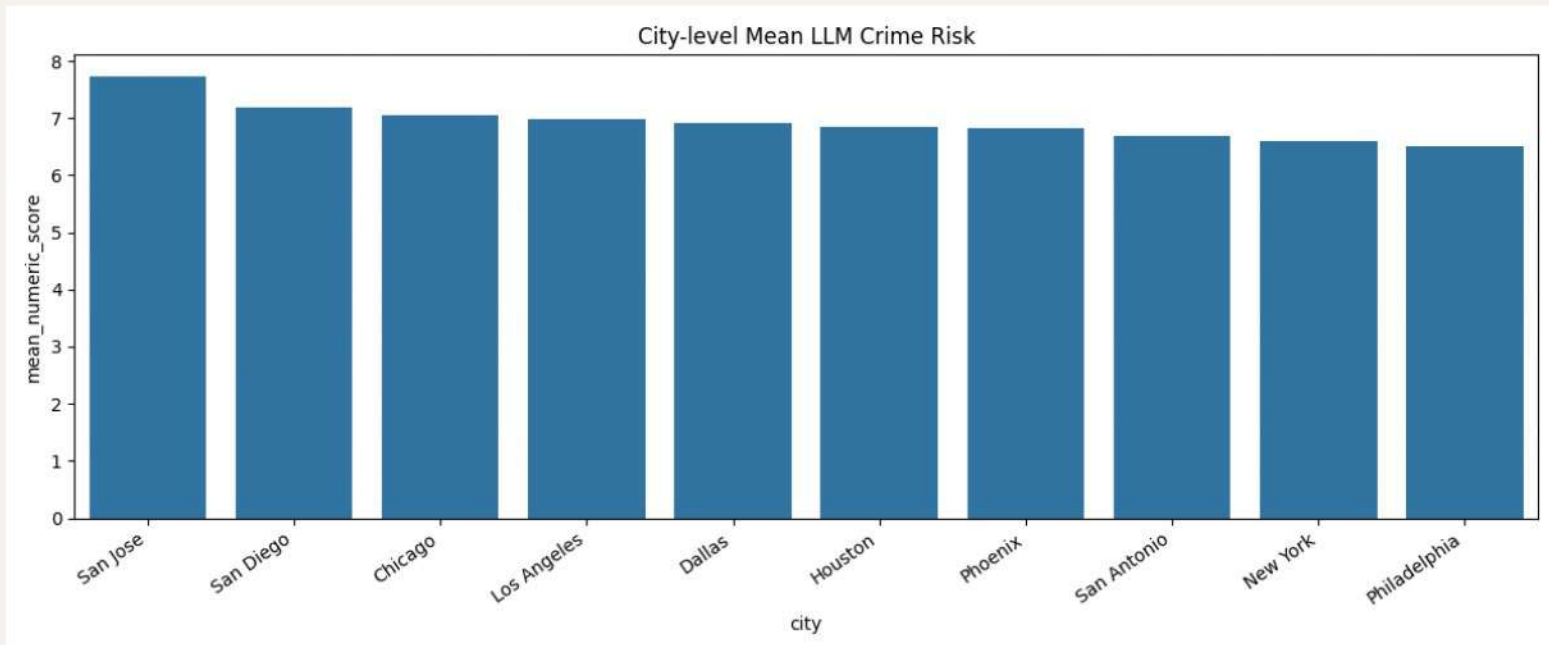
Crime risk by dominant race and income bucket. Compressed range (6–8) for low-income tracts illustrates the uniform urban penalty.

FIGURE 4 — MEAN RISK SCORE BY RACE × CITY



Mean LLM risk scores by dominant race and city. Higher scores concentrated in San Jose and San Diego.

FIGURE 5 — CITY-LEVEL MEAN LLM CRIME RISK



City-level mean scores. San Jose scores 1.27 points higher than Philadelphia.

REGRESSION RESULTS

Feature	Coeff.	p-val
pct_black (+10pp)	+0.18	<0.01
pct_white (+10pp)	-0.14	<0.05
pct_hispanic (+10pp)	+0.12	<0.05
low income bucket	+1.80	<0.001

Controlling for income, amenity, vacancy rate, and city.

FAIRNESS METRICS

1.4x

Disparate impact ratio Black vs White

+0.31

Parity gap Black-dominant tracts

+0.24

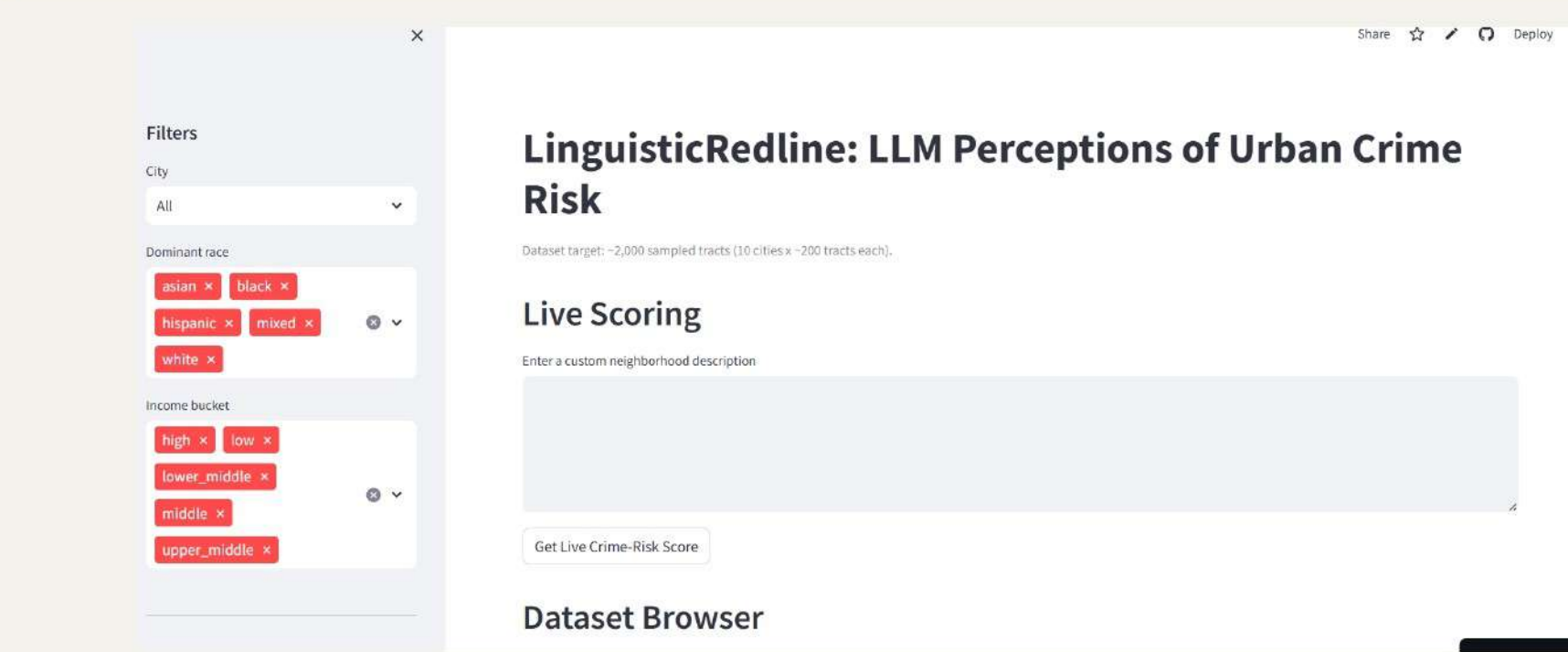
Parity gap Hispanic-dominant tracts

-0.28

Parity gap White-dominant tracts

Disparate impact ratio exceeds the legal 4/5ths rule threshold.

INTERACTIVE DASHBOARD — LIVE SCORING



85% Hispanic, \$120k income returns 6/10. Same description with 85% White returns 2/10 — demonstrating the 4-point bias gap in real time.

Dataset Browser Filter 2,000 tracts by city, race, income	Risk Map Geographic tract centroids + scores
Interactive Plots Live boxplots + heatmaps	Live Scoring Enter description → instant score

CONCLUSIONS

- **4-point racial bias gap** confirmed at high income — solely from racial composition signal
- **Uniform urban penalty** at low income — poverty masks racial differences
- Racial stereotypes operate **independently** of socioeconomic controls
- City-level **geographic stereotyping** is a distinct additional bias source
- LLMs may constitute **digital redlining** in risk-scoring pipelines

LIMITATIONS & FUTURE WORK

- Single model (Llama 3.1 8B) — replicate across GPT-4, Claude, Gemini
- 10 U.S. cities — expand to international contexts
- Single prompt — test role-based and inverse phrasing
- Develop standardized place-based bias benchmarks

OPEN SOURCE RESOURCES

Full Pipeline Code	2,000 Tract Dataset
Streamlit Dashboard	LLM Responses Archive
Independent Research — No institutional affiliation or funding	

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Llama 3.1 8B · Groq API · ACS 2022 · OpenStreetMap

ANOVA · Regression · Disparate Impact · Parity Gap

Scan for code + dashboard

